



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Filter Table

Wavetable Spectral Filter

Turns wavetable frames into filter responses: choose from 120 factory tables or draw your own, morph through 256 frames, and set the phase, contour, and motion.

120 factory tables

256-frame morphing

4 phase modes

20 factory presets



Contents

- | | |
|-------------------------|------------------------------------|
| 01 Overview | 06 Presets |
| 02 Quick start | 07 Recipes |
| 03 Interface | 08 Tips and troubleshooting |
| 04 Controls | 09 Parameter reference |
| 05 Custom tables | 10 Version history |
| | 11 Support and license |

01 Overview

TB Filter Table uses wavetables as filters. Each table is a sequence of single-cycle frames on a 256-slot rail, and the current frame is turned into an FIR filter response that shapes your audio. Moving through the frames morphs the filter, from gentle contours to formants, notches, combs, and hard pulse shapes.

CUTOFF (H24) places the frame on the frequency axis, **RESONANCE** sets the contour depth, **FRAME** and the **MORPH** LFO move through the table, **PHASE** sets the phase response, and **MIX** blends with the latency-compensated dry signal. The EDIT view adds **DRIVE**, the plugin presets, and tools for drawing and capturing your own custom table.

Key features

- 120 factory tables in 10 families, plus an editable CUSTOM table
- 256-slot frame rail; tables morph smoothly or in steps between 2 to 256 keyframes
- CUTOFF (H24) from 20 Hz to 20 kHz, with an effective floor of 200 Hz
- RESONANCE contour depth from flat (−100%) to squared (+100%)
- Four phase modes: MINIMUM, LINEAR, ORIGINAL, and RAW
- Post-filter DRIVE up to 24 dB
- MORPH LFO with six shapes, ±100% range, and a 0.01 to 10 Hz rate
- Custom table drawing, frame capture, and .tbft table files
- 20 factory presets, user presets, undo, and redo

Signal flow



About this manual

This manual describes TB Filter Table 1.1.0 for Windows (VST3). The current macOS build (AU and VST3) is version 1.0.1 and may differ.

02 Quick start

- 1 Insert TB Filter Table on a synth, pad, drum, vocal, or bus track. The Init preset uses the Moving Formant table, **FRAME** 129, **CUTOFF** 4 kHz, **RESONANCE** +15%, and **MIX** 75%.
- 2 Choose a table with the table selector **4**, or step through all tables with its arrows.
- 3 Sweep **FRAME** **6** to morph through the table, and move **CUTOFF** **7** to place the response.
- 4 Set the depth with **RESONANCE** **8** and the balance with **MIX** **9**.
- 5 For movement, choose a **TYPE** in the MORPH LFO row **10**, then set **RANGE** and **RATE**.

03 Interface

The main view holds the everyday controls. **EDIT** **5** opens the secondary tools, and **DONE** **13** returns to the main view.



TB Filter Table 1.1: the main view with the Init preset (left) and the secondary tools with the TABLE TOOLS tab (right).

- 1 **PHASE** — MINIMUM, LINEAR, ORIGINAL, or RAW.
- 2 **Table view** — the frames of the current table, with the current frame highlighted.
- 3 **TABLE FIR RESPONSE** — the realised filter response; the marker shows CUTOFF.
- 4 **Table selector** — CUSTOM and the 120 factory tables, grouped by family; the arrows step through all of them.
- 5 **EDIT** — opens the secondary tools.
- 6 **Frame** — position on the 256-slot rail.
- 7 **Cutoff** — the frequency assigned to harmonic 24.
- 8 **Resonance** — contour depth.
- 9 **Mix** — dry/wet blend.
- 10 **MORPH LFO** — TYPE, RANGE, and RATE of the frame LFO.
- 11 **TABLE** — the table's frames; the rail above shows the 256 slots, the keyframes, and the current frame.

- 12 DRAW FRAME** — the current frame; drag inside it to draw the custom table, then release to commit.
- 13 DONE** — returns to the main view.
- 14 TABLE TOOLS / FRAME LFO** — switches between the table tools and the Frame LFO controls.
- 15 FRAME, DRIVE** — frame position and post-filter drive.
- 16 Keyframe status** — the number of keyframes and the morph mode (SMOOTH or STEPPED).
- 17 CUSTOM TABLE** — NEW, CAPTURE, CLEAR, and TABLE FILE.
- 18 PRESET / HISTORY** — plugin presets, Save, UNDO, and REDO.

04 Controls

Control	Range · default	What it does
Table selector	CUSTOM or 120 factory tables · Moving Formant	Chooses the set of frames the filter morphs through. A new table changes both the tone and the meaning of every frame position.
FRAME	1 to 256 · 129	Base position on the table's 256-slot rail; the filter morphs continuously as it moves. Positive RANGE moves upward from here; negative RANGE moves downward to here.
CUTOFF (H24)	20 Hz to 20 kHz · 4 kHz	The frequency assigned to harmonic 24 of the frame. At 4 kHz the frame spans 20 Hz to 20.48 kHz across ten octaves. The effective cutoff does not go below 200 Hz; lower settings act as 200 Hz.
RESONANCE	−100 to +100% · +15%	Contour depth: −100% is flat, 0% follows the table, and +100% squares it. In RAW mode it sets how far the kernel departs from a neutral one. High values create strong peaks and ringing.
PHASE	MINIMUM, LINEAR, ORIGINAL, RAW · MINIMUM	The phase response of the filter. ORIGINAL keeps the phase of the source frame; RAW uses the frame waveform itself as the filter kernel.
DRIVE (EDIT view)	0 to 24 dB · 2 dB	Post-filter nonlinear color. At 0 dB it is bypassed exactly, without adding a static cut. Drive can raise the level.
MIX	0 to 100% · 75%	Blends the filtered signal with the latency-compensated dry signal.

Morph LFO

Control	Range · default	What it does
TYPE	OFF, SINE, TRI, SAW, SQR, RND · OFF	The frame LFO shape. RND is a random sample-and-hold.
RANGE	−100 to +100% · +25%	Directional morph span: +100% moves up to 256 slots above FRAME, −100% moves up to 256 slots below it, and 0% stops frame movement.
RATE	0.01 to 10 Hz · 0.25 Hz	Free-running LFO rate. It does not follow the host tempo.

In the EDIT view, the FRAME LFO tab shows the same TYPE, RANGE, and RATE controls. TB Filter Table also has a host Bypass parameter, and it reports a fixed latency of 128 samples (about 2.7 ms at 48 kHz) in every phase mode.

The phase modes change how transients and stereo content behave. Compare them on drums, and check the result in mono.

05 Custom tables

The CUSTOM entry at the start of the table list is your own editable table. Build it in the EDIT view with the TABLE TOOLS tab.

Tool	What it does
NEW	Resets the custom table to a new two-keyframe starting table.
CAPTURE	Copies the currently rendered frame into the same slot of the custom table.
CLEAR	Removes the selected authored keyframe from the custom table.
DRAW FRAME	Drag inside the current waveform panel, then release to commit the drawn frame.
TABLE FILE	IMPORT TABLE FILE... and EXPORT CUSTOM TABLE... read and write .tbft table files. Table files are separate from plugin presets.
SMOOTH / STEPPED	SMOOTH interpolates continuously between adjacent keyframes; STEPPED switches between them without interpolation. The keyframe status shows which applies.

06 Presets

Plugin presets store all settings. Choose them in the PRESET / HISTORY row of the EDIT view, or step with its arrows. Save asks for a name and stores a user preset in Documents/TB Filter Table/Presets/User. UNDO and REDO step through editable actions.

Preset	Table	FRAME	CUTOFF	RESONANCE	DRIVE	MIX	PHASE	MORPH LFO
ESSENTIALS								
Init	Moving Formant	129	4 kHz	+15%	2 dB	75%	MINIMUM	OFF
Gentle Pad Motion	Reed Breath	73	1.8 kHz	-18%	0 dB	82%	MINIMUM	OFF
Hollow Drum Focus	Open Notch	117	2.6 kHz	+24%	2.5 dB	72%	MINIMUM	OFF
Vocal Formant Lift	Moving Formant	145	3.4 kHz	+36%	1.5 dB	68%	ORIGINAL	OFF
Subtle Bus Contour	Pink Horizon	49	5.2 kHz	-32%	0 dB	28%	LINEAR	OFF

Preset	Table	FRAME	CUTOFF	RESONANCE	DRIVE	MIX	PHASE	MORPH LFO
MOTION								
Slow Vowel Bloom	Vowel Bloom	1	2.1 kHz	+22%	0 dB	88%	ORIGINAL	SINE 14.1% 0.07 Hz
Liquid Reed	Liquid Reed	117	1.5 kHz	+10%	1 dB	76%	MINIMUM	TRI 37.5% 0.18 Hz
Golden Sweep	Golden Pulse	1	2.4 kHz	+18%	0 dB	92%	LINEAR	SAW 75% 0.1 Hz
Phase Weave	Phase Weave	141	3.1 kHz	+28%	0 dB	84%	ORIGINAL	SQR 37.5% 0.35 Hz
Comb Garden	Comb Garden	65	4.2 kHz	+54%	3 dB	74%	LINEAR	RND 50% 0.75 Hz
CHARACTER								
Warm Ladder	Warm Ladder	83	1.3 kHz	+16%	4 dB	80%	MINIMUM	OFF
Prime Choir	Prime Choir	177	2.8 kHz	+40%	1 dB	86%	LINEAR	OFF
Glass Tide	Glass Tide	133	4.6 kHz	+34%	2 dB	78%	ORIGINAL	OFF
Dense Fold	Single Fold	153	1.9 kHz	+46%	8 dB	70%	MINIMUM	OFF
Pulse Steps	Triangle Terrace	109	3.6 kHz	+58%	4 dB	66%	RAW	OFF
SHOWCASE								
Raw Barcode	Barcode	207	5.1 kHz	+72%	10 dB	78%	RAW	OFF
Mobius Kernel	Mobius Wave	215	3.3 kHz	+62%	12 dB	74%	RAW	OFF
Prism Bloom	T10 Prism	193	6.4 kHz	+66%	5 dB	82%	LINEAR	OFF
Nested Fold	Nested Fold	229	2.7 kHz	+74%	16 dB	62%	MINIMUM	OFF
Broken Speaker	Double Pulse	185	1.7 kHz	+86%	20 dB	56%	RAW	OFF

Factory tables

Family	Tables
Fundamentals	Pure Sine, Velvet Triangle, Rounded Saw, Hollow Square, Soft Ramp, Twin Sine, Golden Pulse, Arc Triangle, Bell Cycle, Tapered Saw, Balanced Steps, Clear Octave
Harmonic Ladders	Warm Ladder, Odd Lantern, Even Glass, Prime Choir, Fibonacci Air, Triple Bloom, Quarter Stack, Alternating Brass, Inverse Bells, Crescent Partials, Hollow Ladder, Full Spectrum
Spectral Contours	Amber Roll-Off, Silver Rise, Pink Horizon, Violet Edge, Moving Formant, Open Notch, Comb Garden, Phase Stair, Odd Tilt, Twin Formants, Soft Ceiling, Cascade Slope
Geometry	Power Petal, Ellipse Arc, Superellipse, Cubic Bow, Sigmoid Silk, Arcsine Halo, Arctangent Lens, Rational Curve, Circle Fold, Diamond Drift, Parabolic Wing, Catenary Loop
Polynomial Orbits	T2 Mirror, T3 Trident, T4 Compass, T5 Star, T6 Rosette, T7 Crown, T8 Lattice, T9 Orbit, T10 Prism, T11 Wheel, T12 Aurora, T13 Helix
Phase Symmetry	Forward Bend, Reverse Bend, Split Phase, Phase Pinch, Skewed Cycle, Dual Axis, Mirror Delay, Quadrature Bloom, Center Pull, Edge Push, Symmetry Break, Phase Weave
Pulse Architecture	Soft Pulse, Three Level, Fine Ladder, Notch Gate, Double Pulse, Stepped Ramp, Sine Stair, Window Gate, Barcode, Triangle Terrace, Sync Pulse, Rounded Square
Fold and Warp	Single Fold, Double Fold, Ribbon Fold, Sine Within, Cubic Fold, Atan Warp, Triangle Cage, Soft Mirror, Nested Fold, Offset Fold, Wave Knot, Fold Cascade

Family	Tables
Organic Motion	Reed Breath, Cello Grain, Vowel Bloom, Woodwind Curl, Glass Tide, Copper Drift, Petal Motion, Liquid Reed, Warm Current, Slow Orbit, Bamboo Arc, Living Chord
Experimental Curves	Linear Chirp, Nested FM, Ring Orbit, Spiral Phase, Rectified Bloom, Wave Packet, Spectral Mirror, Phase Lattice, Logistic Arc, Rational Orbit, Cusp Motion, Mobius Wave

07 Recipes

Starting points for common jobs. Adjust them by ear.

Vowel pad

Slow, talking motion on pads and drones.

Preset Slow Vowel Bloom

RATE 0.05 to 0.2 Hz

- 1 Lower MIX if the vowel motion is too strong.

Drum contour

A moving notch that keeps drum attacks clear.

Preset Hollow Drum Focus

MIX 50 to 75%

- 1 Compare the PHASE modes and keep the one with the cleanest attack.

Bus color

A gentle spectral shape on a group or mix bus.

Preset Subtle Bus Contour

MIX 20 to 35%

- 1 Sweep FRAME slowly to find the most flattering contour.

Deliberate damage

Broken, speaker-like sound design.

Preset Broken Speaker

DRIVE 12 to 20 dB

- 1 Watch the output level; DRIVE adds level.

08 Tips and troubleshooting

Symptom	What to check
CUTOFF has no effect below 200 Hz.	The effective cutoff stops at 200 Hz; settings from 20 to 200 Hz act as 200 Hz.
The sound rings or whistles.	Lower RESONANCE, or choose a smoother table or frame.
Fast morphing sounds rough.	Lower RATE or RANGE. Fast frame movement can create modulation artifacts.
Transients sound soft or smeared.	Compare the PHASE modes and keep the one with the cleanest attack.
It is louder than bypass.	Lower DRIVE or MIX, and match levels before comparing.
The LFO does not follow the tempo.	RATE is free-running. Set it by ear.

09 Parameter reference

Panel label	Host parameter	Range	Default
Table selector	Table	CUSTOM, 120 factory tables	Moving Formant
FRAME	Frame	0 to 255 (shown as 1 to 256)	128 (shown as 129)
CUTOFF (H24)	Cutoff (H24)	20 to 20000 Hz (effective from 200 Hz)	4000 Hz
RESONANCE	Resonance	−100 to +100%	+15%
DRIVE	Drive	0 to 24 dB	2 dB
MIX	Mix	0 to 100%	75%
PHASE	Phase	MINIMUM, LINEAR, ORIGINAL, RAW	MINIMUM
—	Bypass	Off, On	Off
TYPE	Frame LFO Type	OFF, SINE, TRI, SAW, SQR, RND	OFF
RANGE	Frame LFO Range	−100 to +100%	+25%
RATE	Frame LFO Rate	0.01 to 10 Hz	0.25 Hz

10 Version history

Version	Changes
1.1.0 · 2026-08-17	Restored the published 20 Hz to 20 kHz Cutoff automation mapping while keeping the 200 Hz effective floor; every public parameter ID and the state schema are unchanged.
1.0.0 · 2026-08-13	First release: 120 factory tables with 2 to 256 keyframes for morphable spectral filtering; four phase modes, bounded spectral contours, nonlinear drive, and a Frame LFO.

11 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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